

Unit 7a: Atmospheric Moisture, Air Masses & Fronts

Reading: pages 95–104 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. *Relative humidity* is
 - a) the percentage of water vapor the air holds relative to the maximum it can hold at that temperature
 - b) the temperature at which condensation begins
 - c) the total precipitation a region receives in a year
 - d) the total mass of water contained in the air
2. The temperature at which saturated air condenses is called the
 - a) freezing point
 - b) dew point
 - c) lapse rate
 - d) condensation level
3. Which of the following is a *high-level* cloud, formed from ice crystals?
 - a) Stratus
 - b) Nimbostratus
 - c) Cirrus
 - d) Altostratus
4. An air mass classified as *maritime tropical* (mT) is
 - a) dry and cold
 - b) moist and cold
 - c) dry and warm
 - d) moist and warm
5. A *squall line* is
 - a) a line of severe thunderstorms that forms along a cold front
 - b) the boundary marking an occluded front
 - c) a band of fog that forms along a coastline
 - d) the leading edge of a warm front

6. San Francisco's famous summertime fog forms when warm, moist Pacific air blows over the cold California current and cools below its dew point. This is
- a) radiation fog
 - b) advection fog
 - c) steam fog
 - d) upslope fog

Part B · True or False

2 questions × 1 point = 2 points

7. At a stationary front, winds usually blow parallel to the front but in opposite directions.
TRUE / FALSE
8. A warm front is named for the air mass it is pushing away.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Describe what happens along a cold front, and the weather it typically produces.

10. Explain how an *occluded front* forms, and give the order of the air masses from front to back.

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